



YOSI
Exam Focus Group

YOUNG
OPHTHALMOLOGISTS
SOCIETY OF INDIA

YOSI EXAM ACCELERATOR

REVISION CRASH COURSE

IMAGING INTERPRETATION IN THE EXAM

FOR EXAM GOING RESIDENTS


1ST OCTOBER 2026

THURSDAY


8:00 – 10:00 PM


ONLINE WEBINAR

CORNEA		RETINA			SQUINT
					
Pentacam & Sirius Dr. Avani Hariani	AS-OCT and Specular microscopy Dr. Prannesh Ravi	OCT Macula Dr. Sameeksha Agrawal	FFA & ICG Dr. Nidhi Paharia	B-scan Dr. Aswin P.R.	Hess Chart Dr. Smita Kapoor

GLAUCOMA			NEURO-OPHTHALMOLOGY	OCULOPLASTY
				
Visual Fields: Glaucoma & Neuro Dr. Mainak Bhattacharya	OCT RNFL Dr. Russel Da'Cruz	AS-OCT & UBM Dr. Yashas Goyal	VEP and ERG Dr. Deepika CP	CT & MRI Imaging Dr. Samhitha HR

YOSI - Exam Focus Presidential Working Group (PWG)

						
Dr. Samhitha HR	Dr. Alka Kumari	Dr. Avani Hariani	Dr. Apurva Sthapak	Dr. Janki Bhayani	Dr. Nihar Garg	Dr. Smita Kapoor

YOSI - Governing Council

						
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Dr DIVAKANT MISRA
(Mentor)



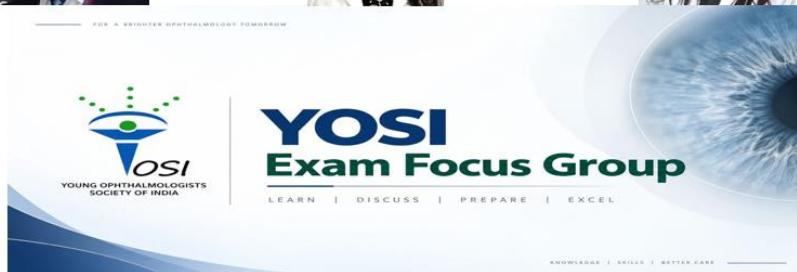
Dr SAMHITHA H R
(Team Lead)



Dr SMITA KAPOOR



Dr NIHAR GARG



Dr AVANI HARIANI



Dr APURVA STHAPAK



Dr JANKI BHAYANI



Dr ALKA KUMARI

MESSAGE

YOSI Exam accelerator is an initiative by the Exam Focus group to help residents overcome their fear of exams and get enough knowledge and pointers before their exams. This PDF has been curated by contributions of all faculty of the “Interpreting Imaging in Exam” webinar. I am grateful to YOSI GC for all the support and encouragement and proud of the work Exam focus group has done in the backdrop to curate this ready reckoner. We hope this helps PG’s in the last minute revision or to serve as a guide while reading a report in clinics.

All the best for exams and clinics.

Compiled by :



Dr Samhitha H R

Team Lead. Exam Focus group

HOD, Oculoplasty ,Orbit, Oculofacial aesthetics

Sr consultant, Cataract, Pediatric & Strabismus

Nethradhama superspeciality eye hospitals, Bangalore

INTERPRETATION OF OCT

Features of an OCT scan

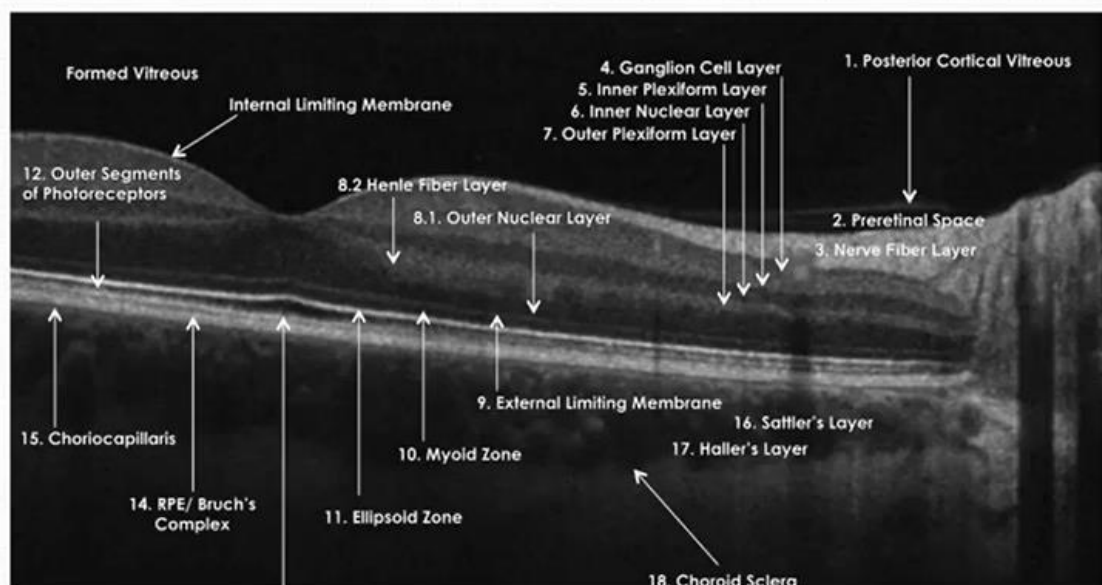
- Patient details
- Date of examination
- Eye
- Scan Position
- Resolution of the scan
- Mode

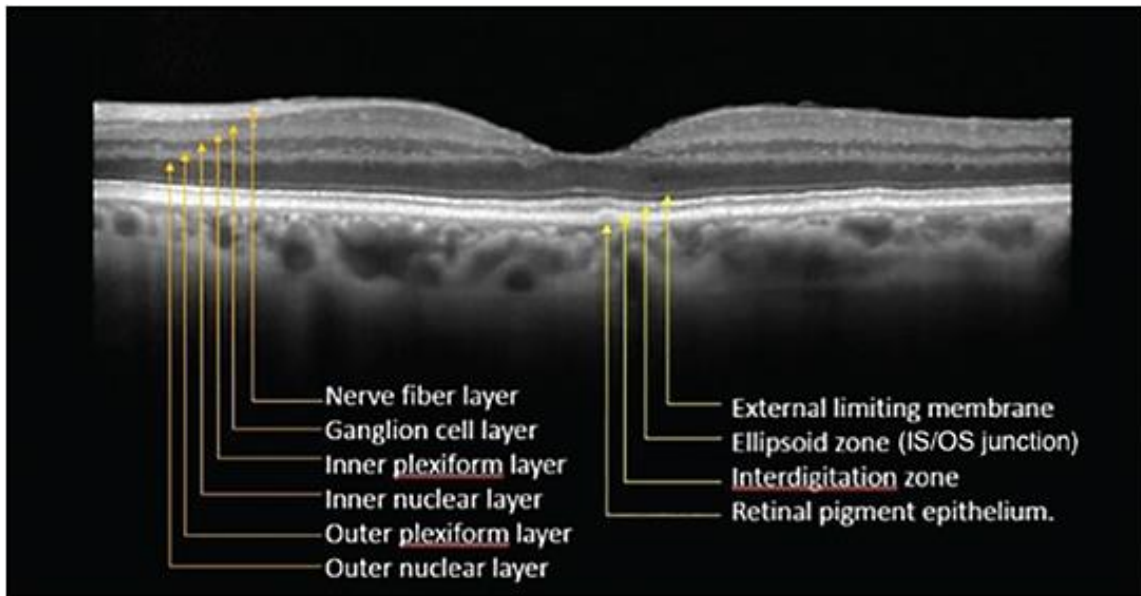
Things seen on normal OCT

- Vitreous
- Interface
- Retinal layers
- Retinal vessels
- Choroid
- Optic nerve head with RNFL layer scan

HYPER REFLECTIVE	HYPO REFLECTIVE
ILM	Ganglionic cell layer
RNFL	Inner nuclear layer
IPL , OPL, ELM, IS-OS junction	Outer nuclear layer
RPE	
RPE- choriocapillaries complex	
Drusen ,PED	Intraretinal/Subretinal –fluid/blood
Epiretinal membrane	
CNVM lesions	
Anterior face of hemorrhage	
Disciform scars	
Hard Exudates	

Normal OCT





1. Start with the basics

OCT modality: SD-OCT / SS-OCT, **Eye:** Right / Left, **Scan quality:** good/adequate/poor, **Signal strength / image quality,** **Scan type:** raster / cube / radial / enhanced-depth imaging, **Scan centration:** fovea-centered or decentered, **Laterality and orientation:** confirm OD/OS and nasal/temporal orientation

Exam phrase:

“This is a good-quality, fovea-centered spectral-domain OCT scan of the right eye.”

2. Assess the vitreoretinal interface

Look systematically from **vitreous** → **retina** → **RPE** → **choroid**.

Vitreous

- Posterior vitreous detachment — complete/incomplete
- Attached posterior hyaloid
- Vitreomacular adhesion — VMA
- Vitreomacular traction — VMT
- Vitreous opacities
- Vitreous hemorrhage

Epiretinal membrane

Look for:

- Hyperreflective membrane on inner retinal surface
- Retinal wrinkling
- Surface corrugation
- Thickening/distortion of foveal contour

- Cystoid spaces
- Loss of normal foveal depression

Important distinction:

VMA = attachment without retinal distortion.

VMT = attachment + anatomic distortion.

ERM = hyperreflective membrane on retinal surface with/without distortion.

3. Assess foveal contour

Ask:

Is the foveal depression normal?

Look for:

- Normal foveal pit
- Blunted foveal contour
- Increased foveal thickness
- Foveal thinning
- Foveal cyst
- Lamellar defect
- Full-thickness macular hole
- Pseudohole

Macular hole — describe precisely

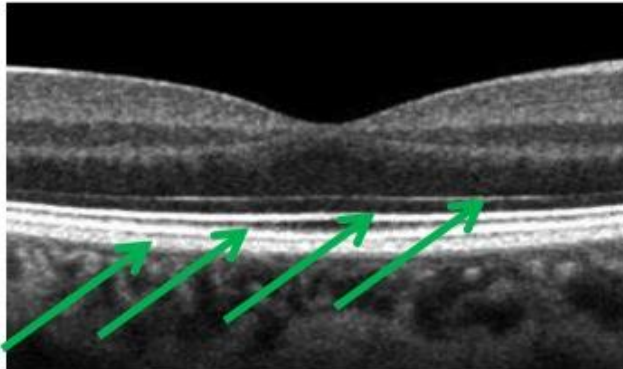
Mention:

- Stage/appearance
- Minimum hole diameter
- Base diameter
- Presence of intraretinal cystic changes
- Operculum
- VMT
- Epiretinal proliferation
- RPE changes

4. Evaluate Retinal Thickness

5. Look for Intraretinal fluid/Subretinal fluid

6. Examine the outer retinal layers



This is increasingly important in retina examinations

Identify: ELM — External Limiting Membrane, EZ — Ellipsoid Zone, Interdigitation zone, Intact/disrupted

Why important?

Outer retinal integrity correlates with photoreceptor function and visual prognosis.

Exam pearl:

“Disruption of the EZ and ELM suggests photoreceptor damage and may have prognostic significance.”

7. Assess the RPE

Look for:

RPE irregularity, RPE elevation, RPE disruption, RPE atrophy, RPE hyperplasia, Drusen, Double-layer sign, RPE tear

8. Examine the choroid

Especially with EDI-OCT.

Assess: Choroidal thickness, Focal thickening/thinning, Choroidal vessels, Haller's layer, Sattler's layer, Choroidal excavation, Pachyvessels

A Simple 10-Point Viva Sequence

Residents can memorize:

“V-F-T-C-I-S-E-R-C-D”

V — Vitreous - PVD, VMA, VMT, ERM

F — Foveal contour - Normal/deformed/hole

T — Thickness - Thick/thin; CMT

C — Cysts - IRF/CME

I — Intraretinal layers - DRIL, hyperreflectivity, thinning

S — Subretinal space- SRF/SHRM

E — EZ/ELM - Intact/disrupted

R — RPE - Drusen, irregularity, PED, atrophy

C — Choroid - Thickness/pachychoroid/lesions

D — Diagnosis - Synthesize the findings rather than naming the disease from one sign.

How to Give the Final Exam Answer

“The OCT shows [key structural abnormalities], involving [specific retinal layers/compartments], with/without intraretinal fluid, subretinal fluid, RPE changes and vitreoretinal interface abnormalities. The outer retinal layers are [intact/disrupted]. The findings are suggestive of _____.”

Then add:

Differential diagnosis

Give 2–3 relevant differentials, if appropriate.

Correlation - “I would correlate with visual acuity, fundus examination and, where indicated, OCTA/FFA/ICGA.”

Management/prognostic relevance - Only if specifically asked.

★ One rule for Ophthalmology practical exams

Do not jump from “cyst” → “DME.”

First describe what you see, then localize it anatomically, then interpret it clinically.

Description → Localization → Interpretation → Differential → Correlation.

Contributed by :



Dr Sameeksha Agrawal

Consultant Vitreo-Retina surgeon

Krishna Eye Foundation, Prayagraj

FFA, ICG

To look for:

What phase?

2. Where is the abnormality?

Disc / macula / vessels / periphery

3. Is it hyper- or hypofluorescent?

4. Why?

Hyper → leakage / staining / pooling / window

Hypo → blockage / nonperfusion

5. What is the diagnosis?

Interpretation

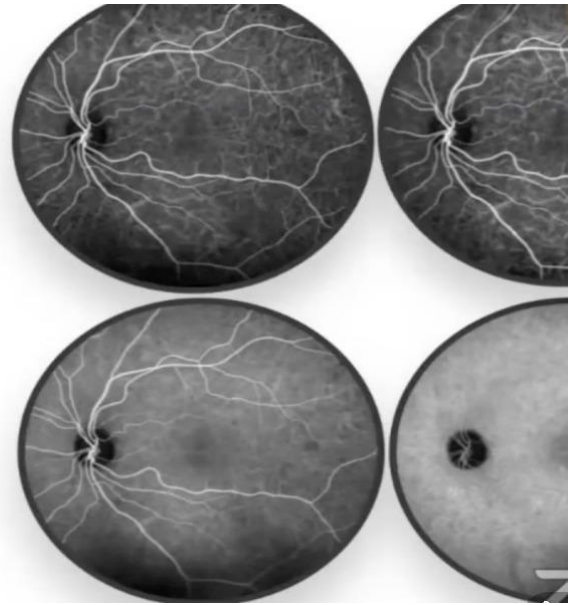
Patient identification

- Eye – centred on disc and macula
- Phase-
- Time sequence -
- Angiographic description
- Level of pathology
- Foveal Avascular Zone
- Probable diagnosis

Time (in sec)	Phase	Event
0	Initiation	Dye Injection
10	<u>Choroidal Flush</u>	Reaches Short Post <u>Ciliary</u> arteries
11-12	Arterial phase	Reaches Central Retinal Artery
13	Capillary phase	Capillaries are <u>perfused</u>
14	AV phase	Laminar flow in veins
16	Venous phase	Lumen of veins fill
25	Maximal fluorescence	<u>Perifoveal</u> capillaries best imaged
30	Recirculation	Removed by kidneys, <u>conc</u> decreases
180-300	Late phase	Leakage , pooling, staining

PHASES OF ICGA

- **Early phase** (1st 60 sec post injection) Choroidal arteries
- **Early mid phase** (1-3 min) Choroidal veins and retinal vessels
- **Late mid phase** (3-15 min) choroidal vessels fade but retinal vessels visible
- **Late phase** (15-45 min) hypofluorescent choroidal vessels & gradual fading of diffuse hyperfluorescence



Contributed by :



Dr Nidhi Paharia

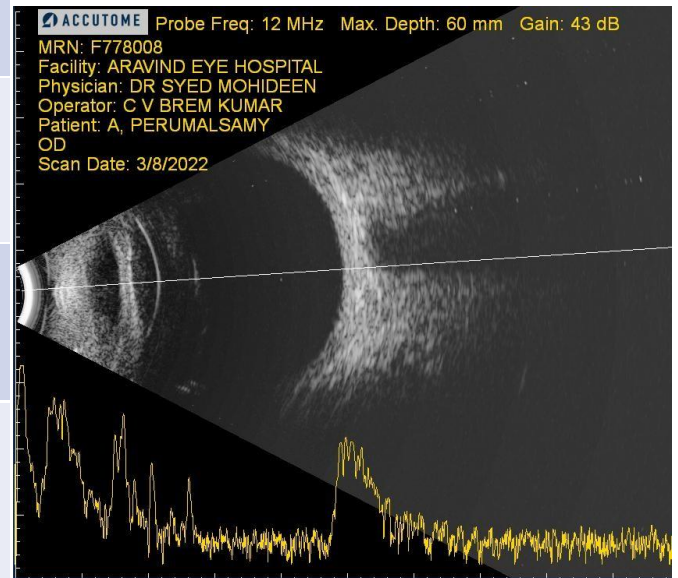
Consultant VR surgeon

Marudhara eye hospital, Jodhpur

Ultrasound B scan

Normal B scan

Lens	Seen with standoff technique or immersion scan, oval high reflective +/- intralesional echoes ~ depending on the amount of cataract
Vitreous	Acoustically clear but can show low echoes depending on the amount of syneresis
RCS complex	Three are seen as a single high echogenic structure.
Optic nerve	Seen as a wedge-shaped acoustic void in the retrobulbar region on an axial scan. Vertical transverse approach at low gain settings is the ideal view
Extraocular muscles	Echolucent to low echogenic fusiform structures within the orbit



Differentiate the dots

	Asteroid Hyalosis	Vitreous hemorrhage	Vitritis
Dot echoes in vitreous	Present	Present	Present
Distribution	Centre	Diffuse, Can settle down in long standing cases	Diffuse

Echo-lucent Zone	Present	Absent	Absent
RCS thickening	Absent	Absent	Present

PVD vs RD vs CD

FEATURES	PVD	RD	CD
SHAPE	V/U	LINEARV/U	DOME
LOCATION	VARIABLE	VARIABLE	PERIPHERY(PREEQUATOR)
ATTACHMENT TO ONH	VARIABLE	YES	NO
OTHER FINDINGS	PROMINENTLY SEEN INFERIORLY	FOLDS,BREAKS,PVR CHANGES	Serous/Hemorrhagic KISSING CHOROIDALS
SPIKE HEIGHT ON A-SCAN	40-90%	80-100%	90-100%
SPIKE PEAK	SINGLE	SINGLE	DOUBLE
MOBILITY	MARKED	MODERATE	MINIMAL
AFTER MOVEMENT	MARKED	MINIMAL	-

Choroidal tumours

	Melanoma	Metastases	Hemangioma
Reflectivity	Low/Medium	Variable	High
Internal Structure	Regular	Irregular	Regular
Sound attenuation	High	Variable	Low

Take home message

Do not jump to the diagnosis - Read the scans, the eye, the Gain settings, Measurements

Describe the lesion, Use the correct term – Echogenicity , Spike height / Amplitude

Keep differentials in mind

Always correlate clinically/ with other findings

Contributed by :



Dr Aswin P R

Consultant – Retina and Uvea

Dr Prem's Eye Hospital, Vanchiyoor, Thiruvananthapuram

PENTACAM & SIRIUS TOMOGRAPHY

A one-night-before-exam reading checklist

READ IN THIS ORDER

- 1. QUALITY & ORIENTATION — Confirm patient and eye; check fixation, centration, coverage and data gaps. Repeat a poor-quality scan. Identify the device and map scale.
- 2. CURVATURE — Hot colours indicate steepness; cool colours indicate flattening. Look for a symmetric bow-tie, asymmetric bow-tie, skewed axes, inferior/superior steepening, or a focal/irregular pattern. Axial (sagittal) maps show overall shape; tangential maps localize curvature changes.
- 3. ELEVATION — Review anterior and posterior maps. Identify the reference surface (BFS/enhanced reference) and scale. A symmetric hourglass may occur with regular astigmatism; skewed hourglass, tongue-like extension, irregular hourglass or isolated island may be suspicious. Elevation is relative to the reference surface—not curvature.
- 4. PACHYMETRY — Note central thickness, thinnest pachymetry (TL) and its location. Compare TL with the apex/pupil centre. Assess the thickness distribution, focal thinning and displacement—not just the central number.
- 5. INTEGRATE — Do steepening, elevation and thinning correspond? Check fellow-eye asymmetry and correlate with refraction, slit lamp, retinoscopy and history. No single index diagnoses or excludes ectasia.

HIGH-YIELD VALUES

- Kmax difference between eyes: <2 D; Kmax–K2 in the same eye: <1 D.
- Thinnest-point thickness difference between eyes: <30 μm ; TL–pachymetry apex difference: normally ≤ 10 μm .
- Thinnest-point Y-coordinate: normal <0.50 mm; suspect 0.50 – 1.00 mm; abnormal >1.00 mm.
- Inferior–superior (I–S) difference: usually <1.5 D. If superior exceeds inferior, S–I is usually <2.5 D.
- Anterior Q: usually -1 to 0 . Positive Q = oblate; Q <-1 = hyperprolate.

PENTACAM — PACHYMETRIC PROGRESSION & BAD-D

- CTSP = thickness progression from the thinnest point towards the periphery; PTI = percentage thickness progression. A quick-slope (curve departs early, before 6 mm) or S-shaped profile may occur in FFKC/ectasia. Flat profile may occur in Fuchs/guttata; inverted profile may occur in PMD.
- BAD-D combines standardized deviations: Df (anterior elevation change, BFS to enhanced reference), Db (posterior elevation change), Dp (pachymetric progression), Dt (thinnest pachymetry), Dy (vertical displacement of TL from apex). Final D is regression-based—not a simple sum.
- BAD deviation: <1.6 SD = normal; 1.6 to <2.6 = suspicious; ≥ 2.6 = pathological. Review the individual components and maps, not just the final D.

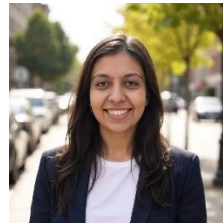
SIRIUS — INDICES & REFRACTIVE ANALYSIS

- Slf / Slb = anterior / posterior symmetry index; positive Slf suggests inferior steepening. Kv_f / Kv_b = anterior / posterior keratoconus vertex. BCV_f / BCV_b / BCV = front / back / total Baiocchi–Calossi–Versaci index (coma/trefoil-related ectatic shape).

- CSIf / CSIb = anterior / posterior centre-surrounding curvature difference. Elf / Elb = anterior / posterior ectasia index. RMS/A = deviation from fitted asphero-toric surface per area; check surface and analysis zone.
- Refractive analysis: MPP = mean pupil power; CYL = corneal cylinder; LSA = longitudinal spherical aberration; LCA = longitudinal comatic aberration; HOA RMS = combined higher-order aberration magnitude. These describe optical performance, not a stand-alone ectasia diagnosis.

EXAM PEARL | Describe the pattern first • quote the numbers second • check whether the maps agree.

Contributed by :

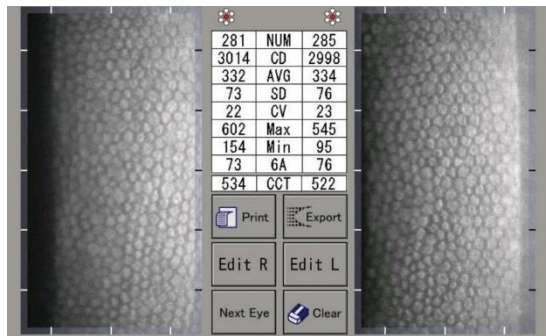


Dr Avani Hariani

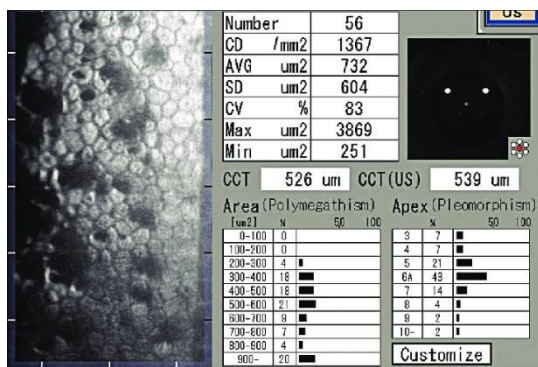
Cornea & Refractive consultant

Shreya eye centre, Delhi

Specular Microscopy

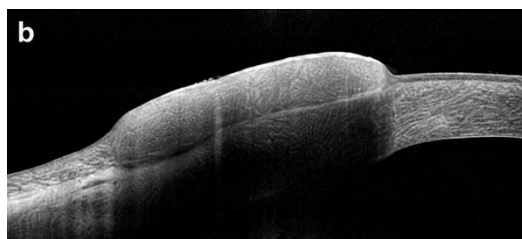


This is a specular microscopy image of the corneal endothelium. Here, the cells in both the eyes show uniform size and predominantly hexagonal morphology. There is absence of polymegathism and pleomorphism. The endothelial cell density is approximately 3000 cells/mm² in both the eyes. The coefficient of variation and hexagonality are within normal limits. Overall, the findings are consistent with normal endothelial morphology, depending on the clinical context."



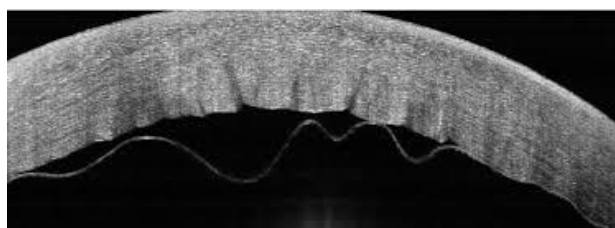
This is a specular microscopy image of the left eye showing significant corneal endothelial morphological abnormalities. The endothelial cell density is 1367 cells/mm², which is reduced, with a mean cell area of 732 μm². There is marked polymegathism, as evidenced by a high coefficient of variation (CV 83%), indicating considerable variation in cell size. There is also pleomorphism, with only 48% hexagonal cells (6A), indicating reduced hexagonality. The central

corneal thickness is 526 μm by specular microscopy and 539 μm by ultrasound pachymetry. Overall, the findings are suggestive of reduced endothelial cell reserve with marked polymegathism and pleomorphism. Clinical correlation encouraged



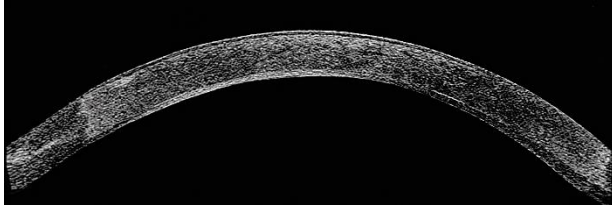
correlated.

This is an AS-OCT demonstrating a localized, markedly thickened and hyperreflective corneal/conjunctival epithelium with a characteristic abrupt transition from normal to abnormal epithelium, suggestive of ocular surface squamous neoplasia (OSSN), to be clinically



These findings are consistent with Descemet's membrane detachment (DMD).

AS OCT shows a hyperreflective posterior membrane corresponding to Descemet's membrane, which is separated from the underlying posterior stroma by a hyporeflective space. The membrane appears irregular and undulating, with areas of separation from the posterior DM, clearly demonstrating the location and extent of the detachment.



Demonstrates a postoperative corneal graft–host interface following *MANUAL* deep anterior lamellar keratoplasty, with the donor stromal tissue apposed to the residual recipient posterior stroma/Descemet’s membrane. The graft–

host interface is seen as a distinct linear hyperreflective plane, with preservation of the posterior corneal contour and no obvious significant interface fluid or separation. The image is consistent with the postoperative appearance of a manually dissected DALK graft.

Contributed by :



Dr Prannesh Ravi

Cornea and Phaco-refractive consultant

Eye Foundation group of hospitals, Pollachi Branch

HESS CHART

- Compare the shape, size & position of the chart
- ❑ Smaller chart – affected eye
- ❑ Equally spaced inner and outer squares – neurogenic
- ❑ Each square 5° - **inner chart measures 15 °, outer chart measures 30°**
- Compare the inner & outer squares
- ❑ Inner square – gross changes, Outer chart – subtle changes
- Identify the underacting & overacting muscles
- ❑ Inwards displacement on chart -underacting muscle,
Outwards displacement on chart- overacting muscle
- Neurogenic strabismus- Proportional spacing between inner & outer chart, follows muscle sequelae
- Restrictive strabismus - Compressed field, does not follow muscle sequelae

Contributed by :



Dr Smita Kapoor

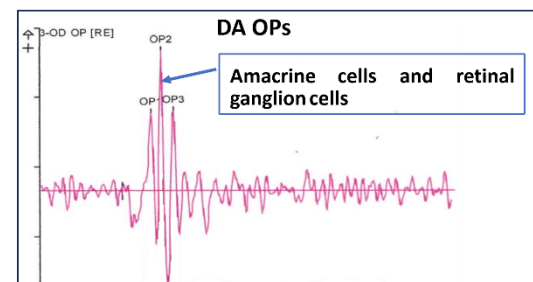
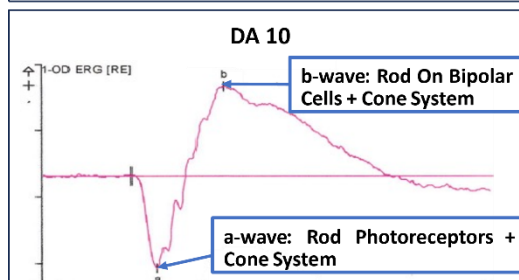
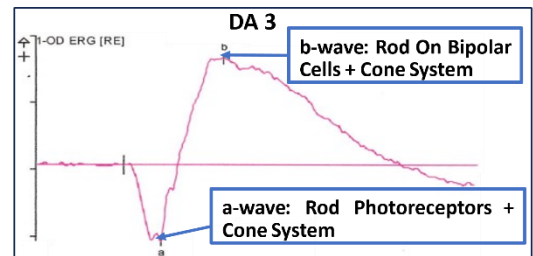
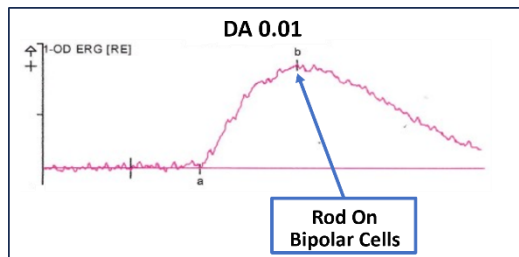
Senior Consultant

*Cataract, Strabismus, Paediatric & Neuro-ophthalmology
Vision Eye Centres, New Delhi*

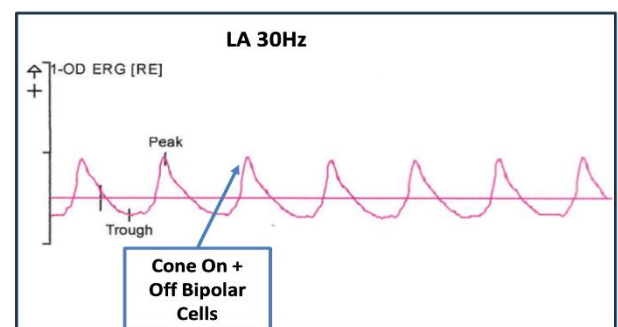
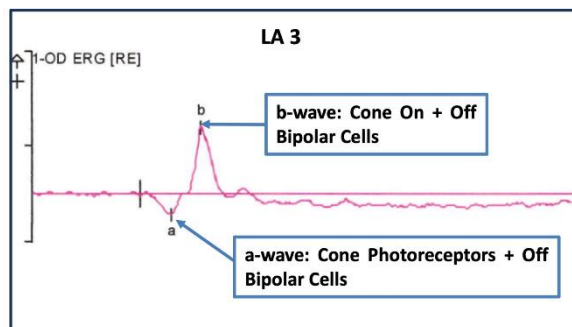
VEP, ERG

1. Full field ERG

Cellular origin of waves : Dark adapted / Scotopic



Cellular Origin of waves :Light adapted or photopic



Interpretation :

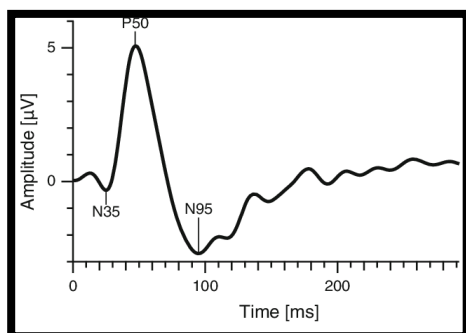
Test Type : Full field ERG				
Eye :				
Condition	ERG Stimulus	Wave	Amplitude	Peak Time

DARK ADAPTED (DA) (Scotopic)	DA 0.01	b	☐ Normal ☐ Reduced ☐ Absent	☐ Normal ☐ Delayed
	DA 3.0	a	☐ Normal ☐ Reduced ☐ Absent	☐ Normal ☐ Delayed
		b	☐ Normal ☐ Reduced ☐ Absent	☐ Normal ☐ Delayed
	DA OPs	OPs	☐ Normal ☐ Reduced ☐ Absent	
	DA 10.0	a	☐ Normal ☐ Reduced ☐ Absent	☐ Normal ☐ Delayed
		b	☐ Normal ☐ Reduced ☐ Absent	☐ Normal ☐ Delayed
DARK ADAPTED (DA) Interpretation : Normal (Generalized rod photoreceptor or bipolar) or Rod dysfunction				
LIGHT ADAPTED (LA) (Photopic)	LA 3.0 Single Flash	a	☐ Normal ☐ Reduced ☐ Absent	☐ Normal ☐ Delayed
		b	☐ Normal ☐ Reduced ☐ Absent	☐ Normal ☐ Delayed
	LA 3.0 Flicker	30-Hz	☐ Normal ☐ Reduced ☐ Absent	☐ Normal ☐ Delayed
LIGHT ADAPTED (DA) Interpretation : Normal or Cone dysfunction (Generalized cone photoreceptor or bipolar)				

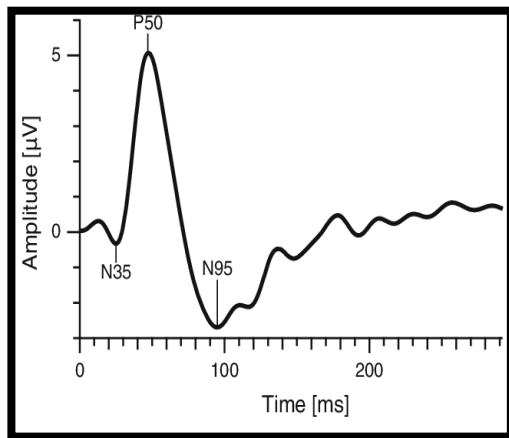
2. Pattern ERG :

- PERG is a localized central retinal response (15 to 30 degree)
- Evoked by a contrast reversing pattern (black and white checker-board)
- Provides information about macular and retinal ganglion cell function
- Differentiation between optic nerve and macular dysfunction

Waveforms : P 50 :



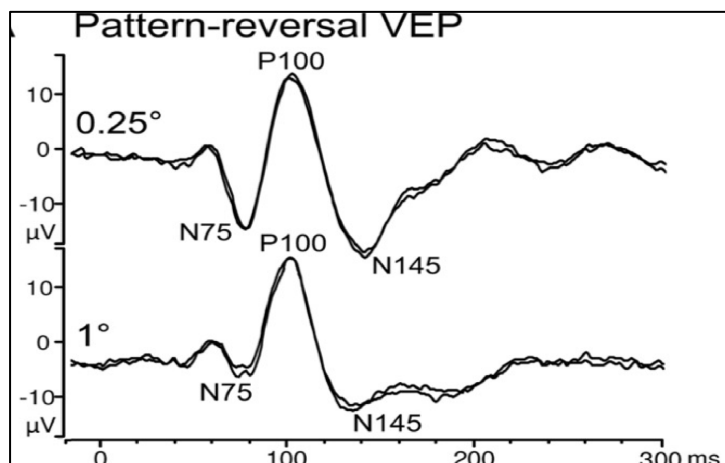
- Initial positive wave
- 75% - Originating from RGCs
- Driven by the cone photoreceptors
- Largely a measure of outer retinal function



- Negative deflection that follows P-50
- Measure of RGCs/ inner retinal function
- Both P50 and N95 depend upon the function of the outer retina and macular cones

3. Pattern VEP :

- Assess the function of primary visual pathway from retina to the primary visual cortex
- Optic nerve and cortical dysfunction
- Affected by macular function, media opacities, patient cooperation



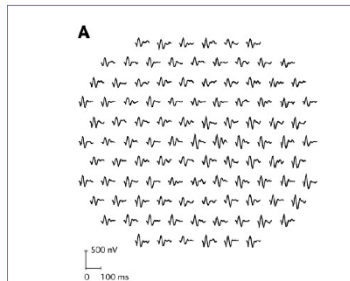
Response elicited by checkerboard stimuli with large 1° and small 0.25° check widths

4. Multifocal ERG

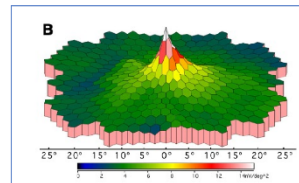
- Localised cone-driven ERG signals recorded from the retina under light-adapted conditions
- Combination of overlapping cone On- and Off-bipolar cell and from cone photoreceptors
- Cone system function in central retina of 40-50 degree

Multifocal ERG

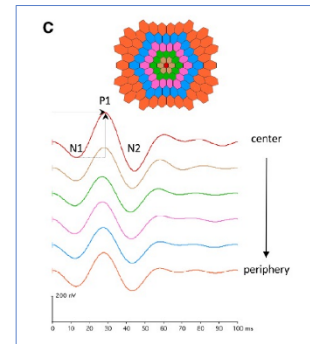
Trace Array



3D-response density plot



Ring-averages



Contributed by :



Dr Deepika C Parameshwarappa

Consultant Ophthalmologist

Department of Vitreoretina, Inherited Retinal

Diseases, Ocular Genetics and Electrophysiology

Narayana Nethralaya, Bengaluru, India

Exam technique

Presenting an OCT RNFL to an examiner

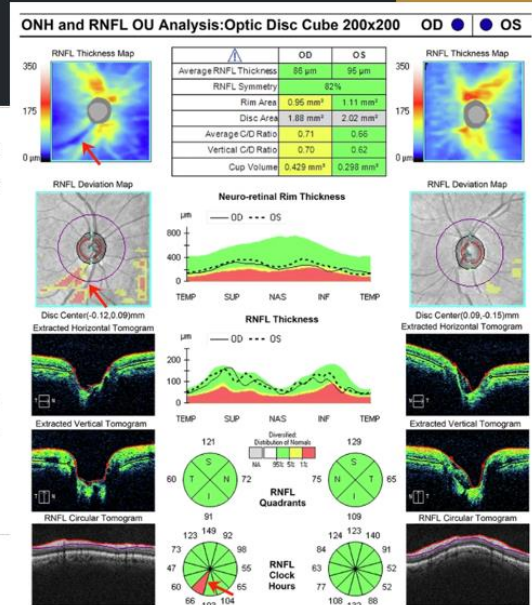
- 1 **Orientate** — state whose eye, which eye, and what scan this is before saying anything about the result.
- 2 **Quality first** — give the signal strength or Q-value out loud, before any color.
- 3 **Describe the global pattern** — average RNFL thickness, and whether the TSNIT profile follows the expected double-hump shape.
- 4 **Go sector by sector, systematically** — quadrants then clock hours, in a fixed order — don't jump straight to the red patch.
- 5 **Compare the two eyes** — state whether the pattern is symmetric or asymmetric before moving on.
- 6 **Correlate** — with the disc photo and visual field if they're provided — an isolated OCT finding is a weaker answer than a concordant one.
- 7 **Conclude with a committed statement** — normal, artifact, or glaucomatous loss — and what you'd do next.

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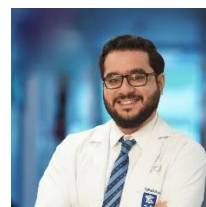
Exam technique

A model answer

“This is an OCT RNFL scan of the right eye. Signal strength is _ out of 10, which is adequate, and there's no evident segmentation error on the B-scan. The average RNFL thickness is at 86 microns. The TSNIT profile shows loss concentrated inferotemporally, which is confirmed as red — below the first percentile — on the deviation map, with the superior quadrant within normal limits. The fellow eye is relatively preserved, so this is an asymmetric pattern. This would correlate with a superior visual field defect, which I'd want to confirm. Overall, this is consistent with glaucomatous RNFL thinning rather than artifact, and I'd want to compare it against prior scans to assess for progression.”



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AS OCT, UBM

1. AS-OCT cannot penetrate behind the pigmented iris epithelium: no ciliary body, no sulcus, no zonules. Plateau iris, iridociliary cysts, IOL in sulcus, malignant glaucoma, cyclodialysis cleft, zonular pathologies need UBM. AS-OCT can capture the scans very quickly (better modality for uncooperative patients), has 360° coverage, and is a non-contact procedure – good for angle imaging, lens vault, bleb morphology, MIGS follow-up.

2. Echogenic, hypoechoic, and echolucent belong to UBM; hyperreflective and hyporefective belong to OCT.

3. For UBM: Radial scan is taken perpendicular to the limbus – used to visualize the angle recess, scleral spur, iris profile, ciliary body profile, sulcus, zonular insertion. In a transverse scan, the beam plane is parallel to the limbus. Once a radial scan has found a lesion, the transverse scan maps its circumferential extent in clock hour. In axial scan, the beam passes through the centre of the cornea and pupil, along the visual axis.

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CT, MRI

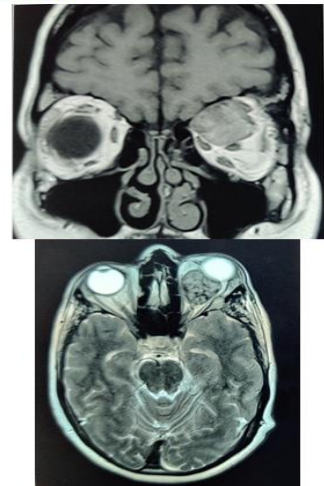
Imaging plane, window, contrast

1. Globes – size, contour, position, internal components
2. EOM
3. Optic nerve and sheath
4. Orbital fat and spaces
5. Lacrimal gland
6. Preseptal area
7. Bony walls and rim
8. Vasculature
9. Paranasal sinuses
10. Brain- apex, cavernous sinus, pituitary, others



Orbital neoplasia

- **Size of the tumour**
- **Circumscription of the tumour**
- **Shape of the tumour**
- **Effect on surrounding structures**
- **Internal consistency**
- **Surrounding bone**
- **Location**
- **Relationship with the adjacent vital structures**
- **Extraorbital extension**



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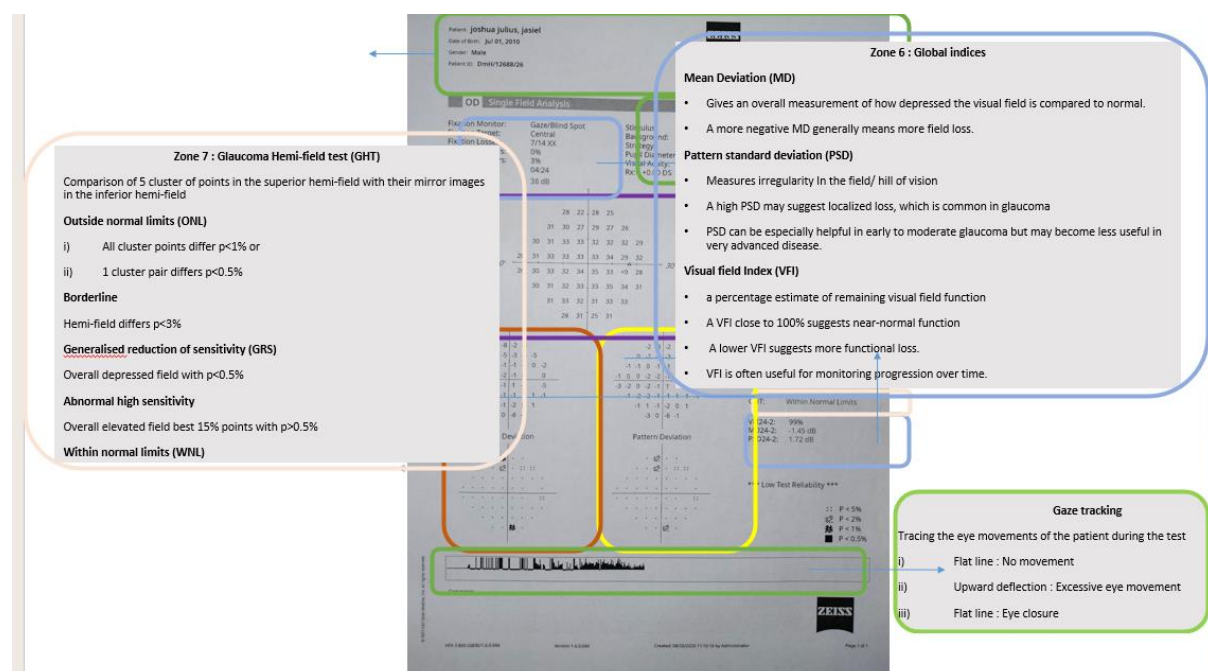
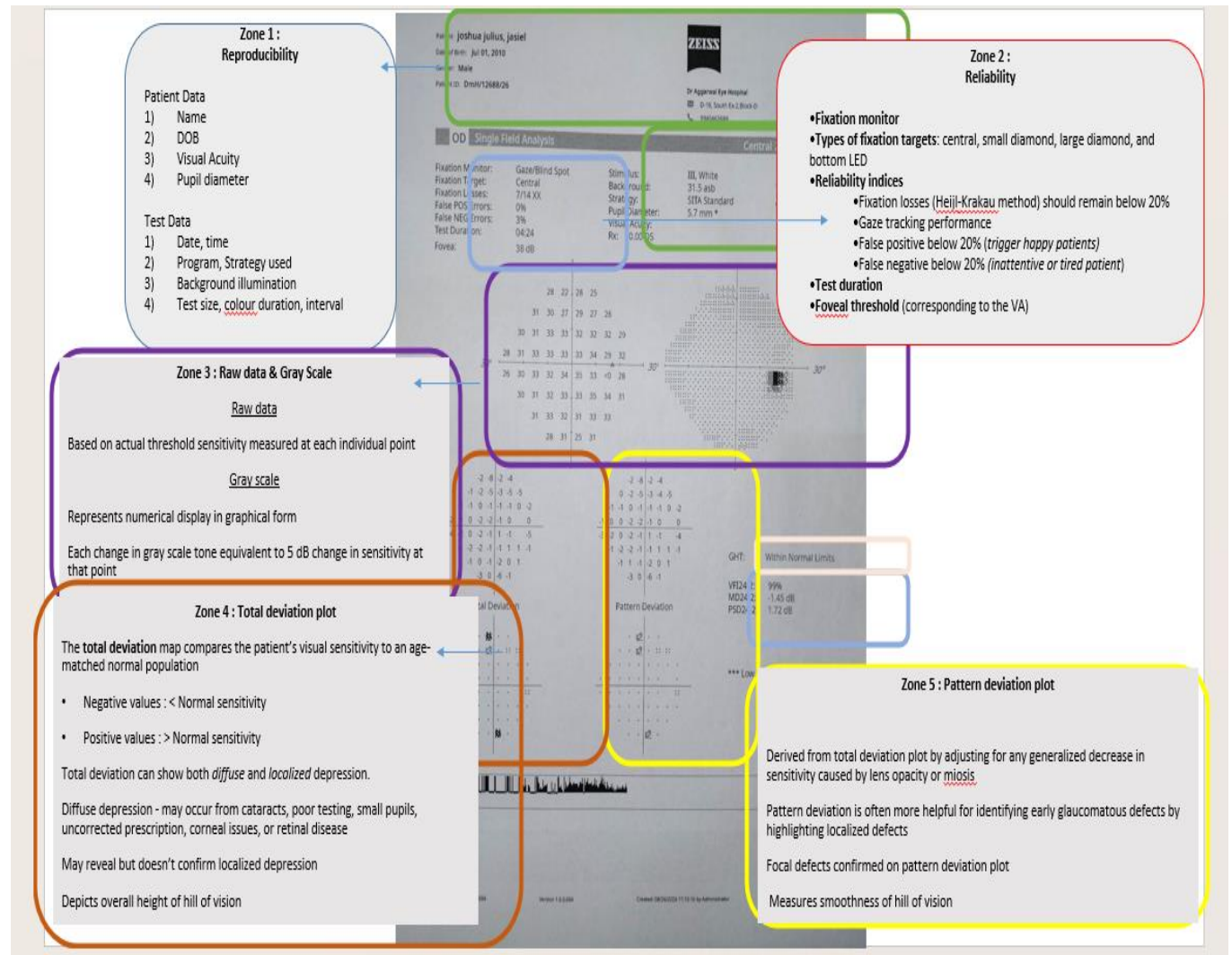
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Visual fields



Key Points:

- Reading a single field print out should always be done stepwise- going sequentially from zone 1 to zone 7 followed by the diagnosis and probable NFL/ONH defect likely to be associated with it
- For OSCE one should be able to identify the pattern by looking at the pattern deviation plot but reliability indices should always be looked at before making a diagnosis
- Anderson's criteria that includes i) 3 or more congruous '*non edge*' points in arcuate area on 30-2 ii) PSD/CPSD $p < 5\%$ and iii) GHT – Outside normal limits should be demonstrable in at least 2 reliable fields
- When comparing serial fields we should look at worsening MD/VFI, expanding scotomas or deepening of existing scotomas, new localized defects that match the optic nerve head changes or RNFL defects
- Non-glaucomatous field defects could be due to test related viz poor test reliability, uncorrected prescription, eyelid obstruction, small pupils, patient fatigue or dry eye or due to other diseases like retinal diseases, non glaucomatous optic neuropathy, neurologic diseases and media opacity like cataract, corneal pathologies.
- For neurologic fields it is mandatory to remember the anatomy of the visual pathway for localising the lesion.
- Lesion may be described as congruous/incongruous, homonymous/heteronymous, central scotoma, ring scotoma, non specific focal depression or generalised constriction

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